

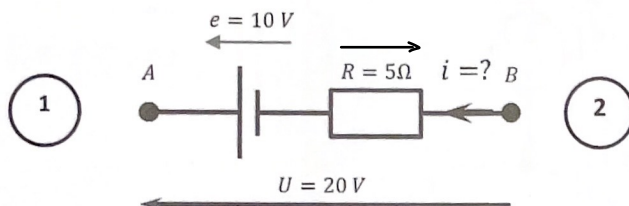
Compétences Visées :

- déterminer les grandeurs dans un circuit

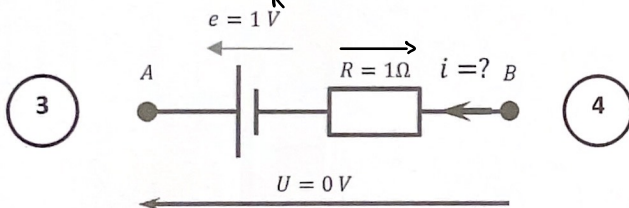
1 EXERCICE D'APPLICATION DIRECTE

Déterminer les grandeurs suivies d'un « ? ».

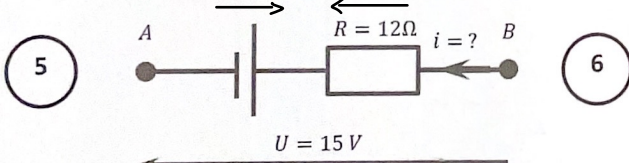
$$U = e - Ri \Leftrightarrow i = \frac{U - e}{-R} \quad \text{AN: } i = \frac{20 - 10}{-5} = -2 \text{ A} \checkmark$$



$$U = e - Ri \Leftrightarrow i = \frac{U - e}{-R} \quad \text{AN: } i = \frac{0 - 1}{-1} = 1 \text{ A} \checkmark$$

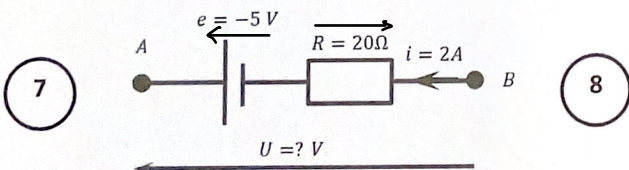


$$-U = e + Ri \Leftrightarrow i = \frac{-U - e}{R} \quad \text{AN: } i = \frac{-15 - 2}{12} = -\frac{17}{12} \text{ A} \checkmark$$



$$U = e - R(-i) \Leftrightarrow R = \frac{U - e}{-i} \Leftrightarrow R = \frac{e - U}{i}$$

$$\text{AN: } R = \frac{4 - 12}{0,8} = \frac{8}{0,8} = 10 \Omega \checkmark$$



$$U = e - Ri \quad \text{AN: } U = -5 - 20 \times 2 = -45 \text{ V} \checkmark$$

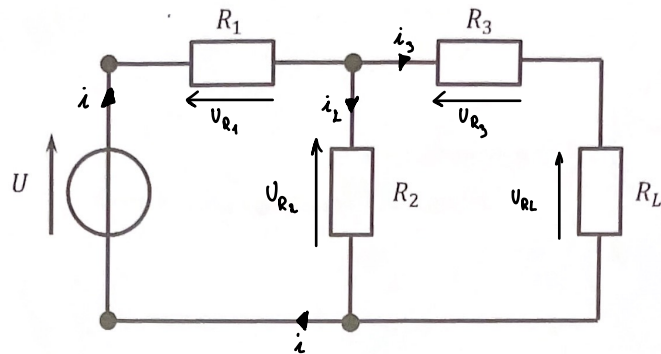
$$U = -e - Ri \Leftrightarrow -e = U + Ri \Leftrightarrow e = -U - Ri$$

$$\text{AN: } e = -15 - 25 \times (-3) = 60 \text{ V} \checkmark$$

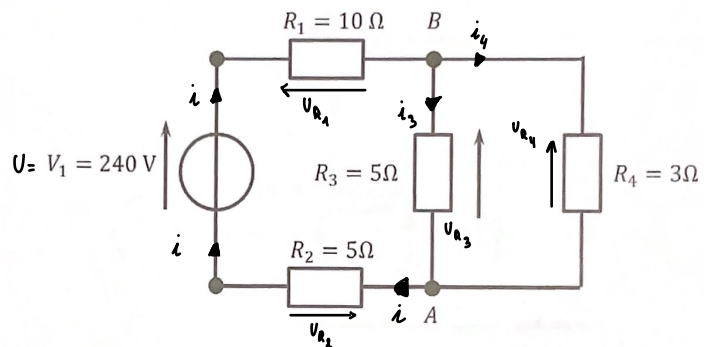
2 ÉTUDE DE CIRCUIT ELECTRIQUE

Dans les circuits suivants, déterminer les intensités en chaque point du circuit ainsi que les différentes tensions.

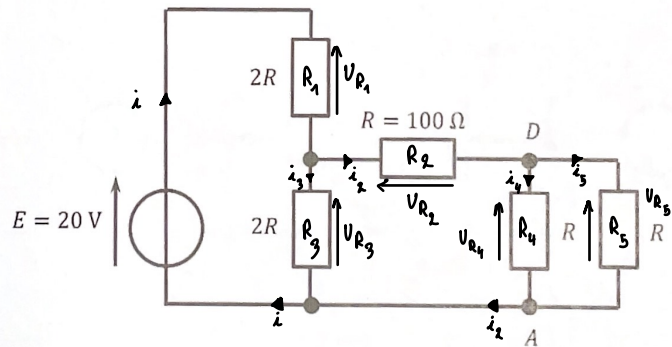
1



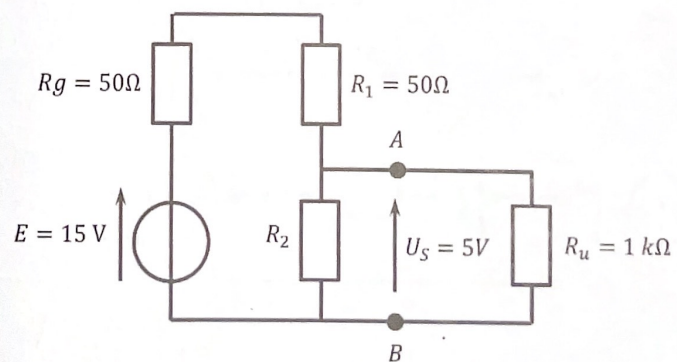
2



3



4



Circuit 2:

$$R_{eq} = R_1 + R_2 + \frac{R_3 R_4}{R_3 + R_4} \quad \underline{AN:} \quad R_{eq} = 10 + 5 + \frac{5 \times 3}{5 + 3} = 17 \Omega$$

$$U = R i \Leftrightarrow V_1 = R_{eq} \times i \Leftrightarrow i = \frac{V_1}{R_{eq}} \quad \underline{AN:} \quad i = \frac{240}{17} = 14 A$$

$$U_{R_1} = R_1 i \quad \underline{AN:} \quad U_{R_1} = 10 \times 14 = 140 V$$

$$U_{R_2} = R_2 i \quad \underline{AN:} \quad U_{R_2} = 5 \times 14 = 70 V$$

$$V_1 = U_{R_1} + U_{R_2} + U_{R_3} \Leftrightarrow U_{R_3} = V_1 - U_{R_1} - U_{R_2} \quad \underline{AN:} \quad U_{R_3} = 240 - 140 - 70 = 30 V$$

$$U_{R_3} = R_3 i_3 \Leftrightarrow i_3 = \frac{U_{R_3}}{R_3} \quad \underline{AN:} \quad i_3 = \frac{30}{5} = 6 A$$

$$i = i_4 + i_3 \Leftrightarrow i_4 = i - i_3 \quad \underline{AN:} \quad i_4 = 14 - 6 = 8 A$$

$$U_{R_4} = R_4 i_4 \quad \underline{AN:} \quad U_{R_4} = 3 \times 8 = 24 V$$

Circuit 3:

$$R_{eq} = R_1 + R_3 + R_2 \parallel (R_4 \parallel R_5) \Leftrightarrow R_1 + R_3 + R_2 \parallel \left(\frac{R_4 R_5}{R_4 + R_5} \right) \Leftrightarrow R_1 + R_3 + \frac{\frac{R_4 R_5}{R_4 + R_5} \times R_2}{\frac{R_4 R_5}{R_4 + R_5} + R_2} \Leftrightarrow R_1 + R_3 + \frac{R_2 R_4 R_5}{R_2 R_4 + R_2 R_5 + R_4 R_5}$$

$$\underline{AN:} \quad R_{eq} = 2R + 2R + \frac{R^3}{R^2 + R^2 + R^2} = 4R + \frac{R^3}{3R^2} = 4R + \frac{R}{3} = \frac{12R + R}{3} = \frac{13}{3} R$$